UNIT
12

# INFORMATION HANDLING

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Information handling is a part of statistics.

## STATISTICS

Statistics is a branch of mathematics concerned with collecting, classifying, analyzing and interpreting.

## FREQUENCY DISTRIBUTION

**Frequency:**
In the raw data, many observations (values or numerical facts) recur again and again. The number of times a value occurs in the data is called its Frequency, usually denoted by 'f'.

**Frequency Distribution:**
A tabular arrangement of data together with class frequencies is called a Frequency Distribution or Frequency Table.

**Construction of a Frequency Table:**
In order to construct a Frequency table, we proceed as under:
**Step 1:** Find the range of the raw-data and determine the class size.
Size of the class interval = (Largest value – Smallest value) / (Number of classes)
**Step 2:** Write down the class-Intervals of equal size.
**Step 3:** Draw an outline of table having columns as under.
**1.** Class-Intervals. **2.** Tally Marks. **3.** Frequency.
Only class-intervals will be mentioned. Other columns will be empty.

| Class Interval | Tally Marks | Frequency |
|---|---|---|
| 45 - 51 | | |
| 52 - 58 | | |
| 59 - 65 | | |
| 66 - 75 | | |

**Step 4:** Find the frequency using tally method.
**Tally Marking:** We tick the members one by one. For each tick we put a tally mark (vertical line-segment) in the Tally Marks column against the corresponding class-interval to which the ticked members belong as explained in solving example 1, given on page 137.

| Class Interval | Tally Marks | Frequency |
|---|---|---|
| 45 - 51 | ~~\|\|\|\|~~ \|\|\|\| | |
| 52 - 58 | \|\|\|\| | |

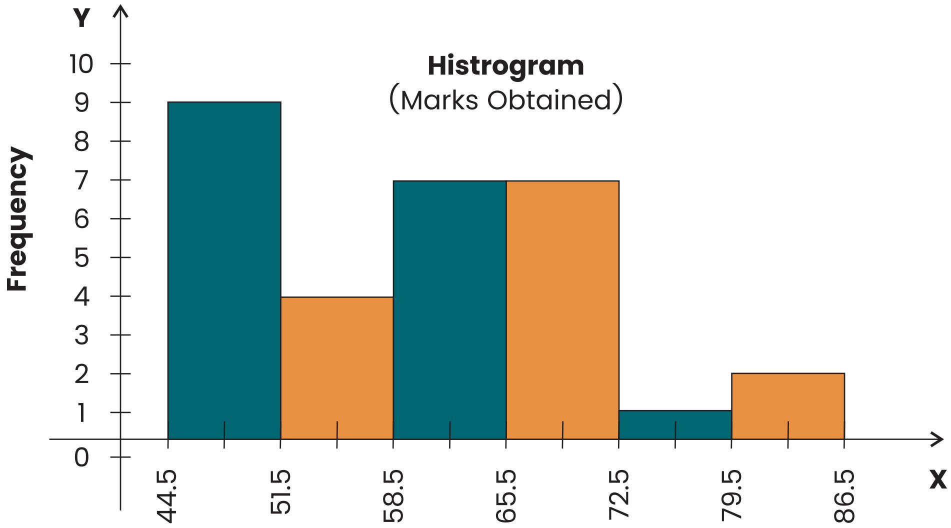
| | | |
|---|---|---|
| 59 - 65 | 𝍸 \|\| | |
| 66 - 75 | 𝍸 \|\| | |

**Example 1.** The following are the marks out of 100 obtained by 27 students of a school in mathematics Test.
67, 44, 68, 56, 69, 50, 65, 51, 66, 51, 53, 54, 65, 53, 48, 54, 63, 62,
61, 59, 67, 45, 68, 50, 69, 50, 65.
Make a frequency distribution table taking 4 class-intervals of equal size.
**Solution.**
Size of class interval = (69 - 44) / 4 = 25 / 4 = 6.25 (So, we take 7).
Thus, 4 class intervals are 45 – 51, 52 – 58, 59 – 65 and 66 – 72.

| Class Interval | Tally Marks | Frequency |
|---|---|---|
| 45 - 50 | 𝍸 \|\|\|\| | 9 |
| 51 - 57 | \|\|\|\| | 4 |
| 58 - 64 | 𝍸 \|\| | 7 |
| 65 - 71 | 𝍸 \|\| | 7 |
| | **Total** | **27** |

**Construct a Histogram Representing Frequency Table.**
To construct a histogram from a grouped data the following steps are followed:
**(i)** Draw X-axis and Y-axis.
**(ii)** Mark class boundaries of the classes along X-axis.
**(iii)** Mark frequencies along Y-axis
**(iv)** Draw a rectangle bar for each class so that the height of the bar drawn for each class is equal to the frequency of the class.
**For example,** the data given below is shown in frequency table.

| Class Boundaries | Frequency | Mid Point x |
|---|---|---|
| 44.5 - 51.5 | 9 | 48 |
| 51.5 - 58.5 | 4 | 55 |
| 58.5 - 65.5 | 7 | 62 |
| 65.5 - 72.5 | 7 | 69 |
| 72.5 - 79.5 | 1 | 76 |
| 79.5 - 86.5 | 2 | 83 |
| **Total** | **30** | |




**Example 2.** 26 Students of class VIII in a school scored the following marks out of 20 marks in a test in Mathematics.

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 10 | 12 | 16 | 2 | 8 | 14 | 16 | 18 | 20 | 13 | 4 | 12 | 6 | 8 | 10 | 14 | 16 | 15 | 16 |
| 16 | 14 | 8 | 12 | 10 | 11 | | | | | | | | | | | | | | |

Suppose the students are graded as:

Marks 16-20, **Grade-A** Marks 11-15, **Grade-B**

Marks 06-10, **Grade-C** Marks 01-05, **Grade-D**

Draw a Histogram and answer the following questions.

**(a)** How many students got Grade-A?

**(b)** How many students got Grade-B?

**(c)** Which grade was obtained by maximum number of students?

**Solution:**

Here the data is not grouped. Thus we have to construct the frequency Table and find class boundaries (Class-Intervals are already given to us). The table is as under:

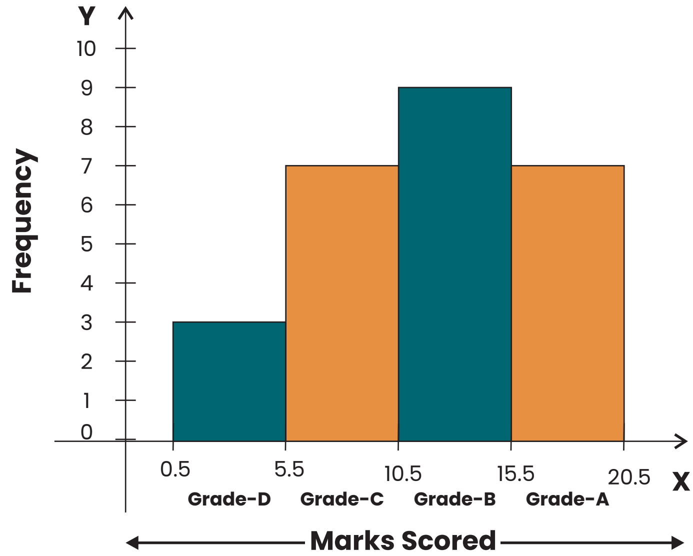
| Class Interval | Tally Marks | Frequency | Class Boundaries |
|---|---|---|---|
| 1 - 5 | \|\|\| | 3 | 0.5 - 5.5 |
| 6 -10 | ~~\|\|\|\|~~ \|\| | 7 | 5.5 - 10.5 |
| 11 - 15 | ~~\|\|\|\|~~ \|\|\|\| | 9 | 10.5 - 15.5 |
| 16 - 20 | ~~\|\|\|\|~~ \|\| | 7 | 15.5 - 20.5 |

We construct the rectsngles following the steps and get the histogram Height of the rectangle indicate that:

**(a)** 7 students got Grade A

**(b)** 9 students got Grade B

**(c)** Maximum number of students got Grade B.




## EXERCISE 12.1

**Q1.** Consider the weights in Kg of 30 boy scouts selected from different schools: 39, 40, 36, 32, 38, 40, 42, 35, 41, 29, 40, 42, 43, 39, 37, 35, 33, 31, 32, 34, 36, 42, 45, 43, 45, 47, 42, 44, 46, 40.
Make a frequency distribution table taking 5 class intervals of equal size by tally marks. Construct its Histogram.

**Q2.** The number of units of electricity consumed by 25 households of a colony are listed below. Construct a frequency table with 5 classes of equal size and then draw a histogram.
700, 720, 750, 740, 750, 730, 690, 695, 690, 695, 700, 710, 705, 690, 685, 695, 690, 705, 710, 700, 680, 677, 720, 725, 730.

**Q3.** Given below are the marks secured by 28 students of class VIII in a monthly test in Mathematics:
60, 58, 62, 61, 59, 65, 67, 67, 64, 60, 63, 70, 67, 69, 70, 80, 79, 78, 71, 82, 50, 45, 55, 56, 52, 54, 59, 60.
Prepare its Frequency Table with 5 classes of equal size and then construct a histogram.

**Q4.** Prepare a frequency Table with 4 classes of equal size using tally method from the data of rainfall (in cm) for the last 25 years. 15, 25, 16, 18, 9, 5, 7, 14, 21, 23, 4, 11, 22, 3, 5, 5, 12, 25, 13, 17, 5, 26, 7, 9, 24. Construct its Histogram.

**Q5.** The following frequency table represents the scores obtained by a group of commerce students in the computer laboratory.

| Marks | 26 - 30 | 31 - 35 | 36 - 40 | 41 - 45 | 46 - 50 | 51 - 55 |
|---|---|---|---|---|---|---|
| No. of Students | 05 | 17 | 13 | 24 | 15 | 14 |

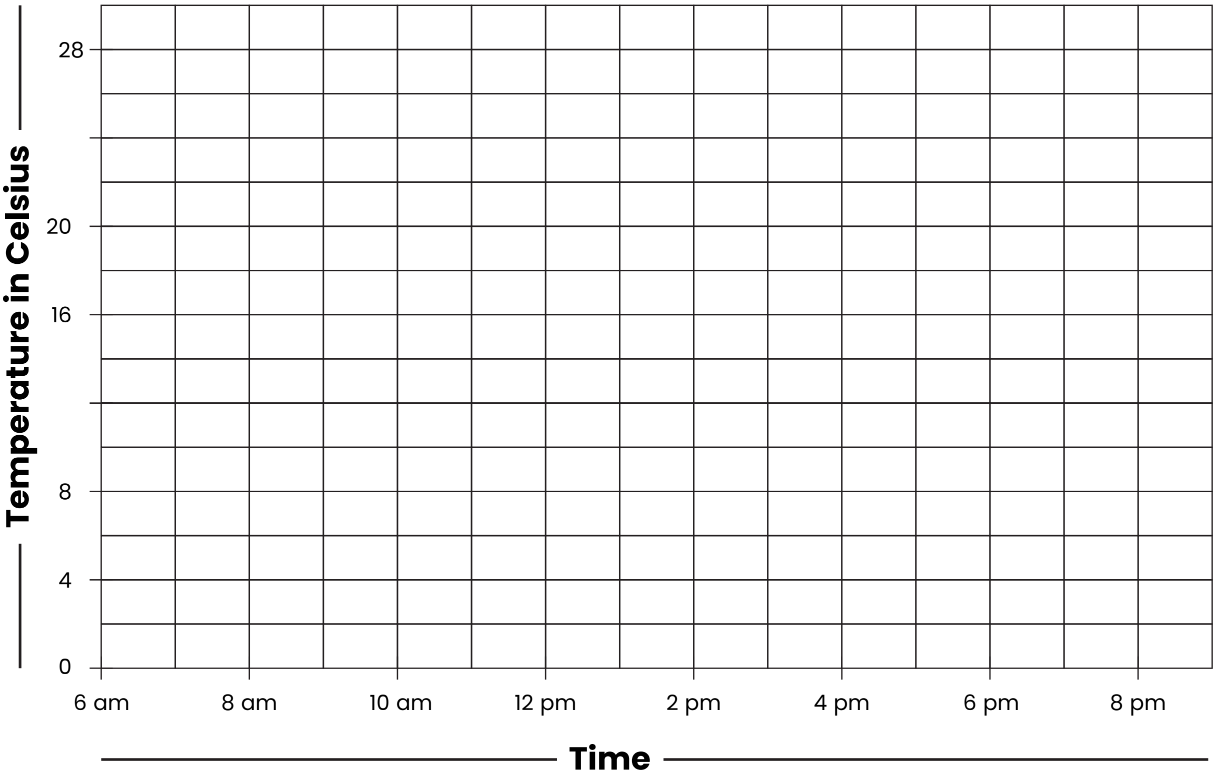
Draw a Histogram for the above distribution.

**Q6.** Construct a Histogram for the distribution of daily earnings in rupees of workers given below.

| Earnings | 501 - 600 | 601 - 700 | 701 - 800 | 801 - 900 | 901 - 1000 | 1001 - 1100 |
|---|---|---|---|---|---|---|
| No. of Workers | 10 | 15 | 20 | 25 | 30 | 35 |

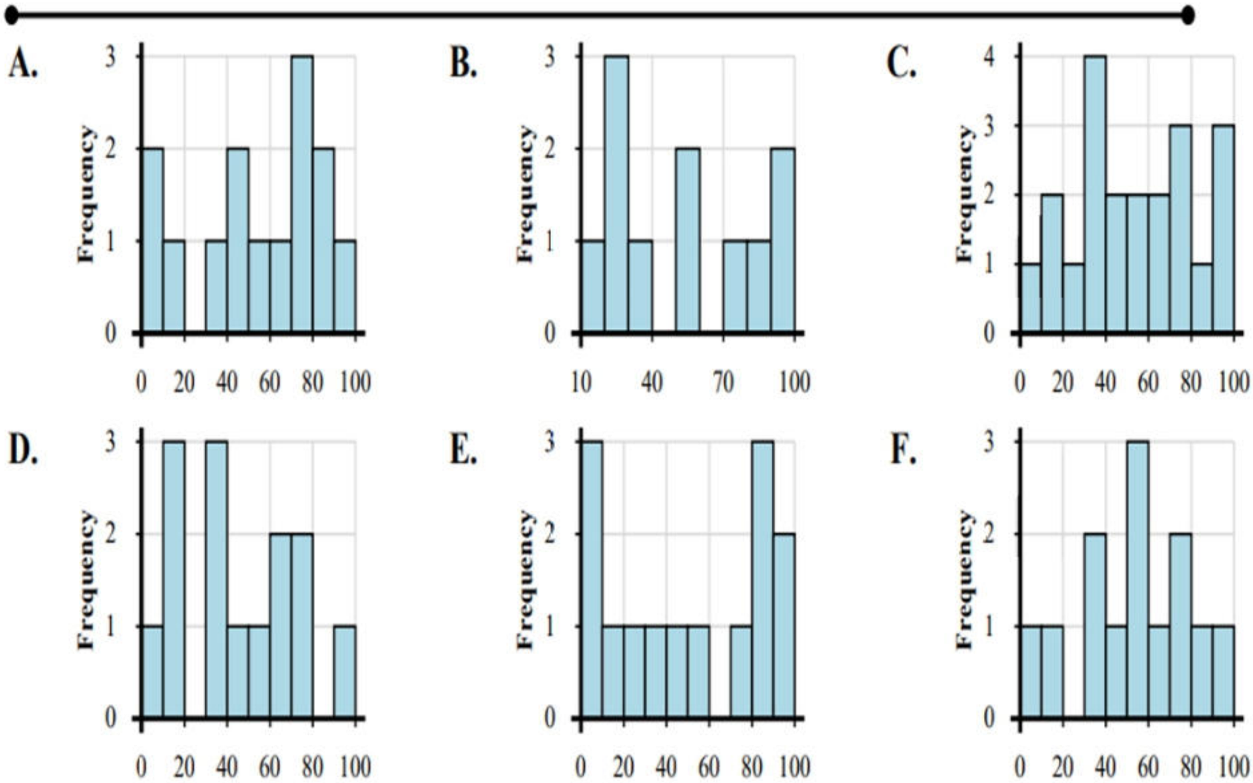
**Q7.** Use the data tables to make histogram.

| 10 C | 12 C | 16 C | 18 C | 22 C | 28 C | 24 C | 18 C |
|---|---|---|---|---|---|---|---|
| 6 am | 8 am | 10 am | 12 pm | 2 pm | 4 pm | 6 pm | 8 pm |



**Q. Match each data set to its histogram.**

**1)** 4, 10, 17, 35, 26, 35, 38, 36, 50, 40, 45, 50, 63, 78, 73, 64, 73, 96, 83, 92
**2)** 6, 10, 13, 10, 35, 33, 35, 40, 55, 64, 72, 78, 63, 91
**3)** 4, 10, 23, 33, 34, 31, 49, 61, 74, 69, 76, 77, 99, 90, 95
**4)** 19, 1, 8, 33, 41, 58, 40, 77, 70, 74, 63, 97, 82, 82
**5)** 5, 18, 5, 9, 31, 23, 59, 42, 73, 82, 99, 85, 84, 89
**6)** 7, 21, 39, 39, 47, 56, 43, 72, 60, 67, 99, 98, 98, 92, 94




**Measures of Central Tendency:**
When the raw data is converted into a frequency distribution. The information given in the data can be further condensed to a single representative value for the entire distribution. As this value lies in the central part of a data, so it is called measure of central tendency.

**Describe measures of Central Tendency:**
The most commonly used measures of central tendency are:
**Mean**
Weighted Mean
Median
Mode

**Mean (Average):**
Mean is said to be the most popular and ideal average. For example, the mean temperature of a city remained 39°C in the month of May means

that the temperature will be above 39°C for some days and it will be below 39°C for other days of May.
The formula for mean is as under.

$$\text{Mean} = \frac{\text{Sum of all the measures}}{\text{number of measures}}$$

or $$X = \frac{x1 + x2 + x3 + \cdots + xn}{n}$$

or $$X = \frac{\text{Sum of the values of all observations}}{\text{Number of observations}}$$

**Note:** Mean is denoted by X and sum by Σ. Hence $X = \frac{\sum_{i=1}^{n} x_i}{n}$

**Example 1.** A student obtained 84, 90, 72, 60, 74, 50, 40 and 52 marks in 8 different subjects in annual examination of class VIII. Find his mean marks.

**Solution:** $\text{Mean} = \frac{\text{Sum of marks in all subjects}}{\text{Number of Subjects}} = \frac{\sum x_i}{n}$

Or $\bar{X} = \frac{84+90+72+60+74+50+40+52}{8} = \frac{512}{8} = 64$ marks.

**Example 2 (i).** If Mean = 6 and $\sum x = 18$, then find n.
**Solution:** Since $X = \sum x / n \Rightarrow n = \sum x / X = 18 / 6 = 3$

**Example 2 (ii).** If X = 4 and n = 5, find $\sum x$.
**Solution:** Since $X = \sum x / n \Rightarrow \sum x = n\,X = 5(4) = 20$.

**Example 2 (iii).** If X = 6, n = 4 and three terms in the data are 2, 8, 10, then find the fourth missing term.
**Solution:** We are given X = 6, n=4 and $\sum x = 2 + 8 + 10 + x_4$
We know that: $X = \sum x / n \Rightarrow \sum x = n\,X \Rightarrow 20 + x_4 = 4\,(6) = 24$
$\Rightarrow x_4 = 24 - 20 = 4$. Hence the fourth term is 4.

**Example 2 (iv).** The mean age of the students of class VIII is 13 years. If the age of Maths teacher is also added, the mean age rises to 14 years. If number of students in the class is 30, find the age of the teacher.
**Solution:** Mean age of students = $X_s = 13$, $n_s = 30$.
∴ Sum of ages of students = $\sum x = n_s \times X_s = 30 \times (13) = 390$ years.
Now Sum of ages of Students + Age of Teacher
$n_{st} \times X_{st} = 31 \times 14 = 434$ years.

Where $n_s$ denotes number of students and $X_{st}$ denotes mean age of students and teacher together. ∴ Age of Teacher = 434 − 390 = 44 years.

### Weighted Mean:

When all the values of the given data have same importance, then we use the mean. However, when different values have different importance then these values are known as weights.

**For example (i)** If Mathematics has more importance than History and English has more importance than Mathematics then we say:
$W_{eng} > W_{maths} > W_{History}$.

**(ii)** Vote of chairman has more importance than vote of a member.

$$\text{Weighted Mean} = \frac{x_1.w_1+x_2.w_2+x_3.w_3+\ldots+x_n.w_n}{w_1+w_2+w_3+w_4+\ldots+w_n} \Rightarrow \bar{X}_w = \frac{\Sigma x_i\, w_i}{\Sigma w_i}$$

**Example 3.** Students of class VIII purchased books of different subjects as under:

| Title of the book | Maths | Science | English | Urdu | Social Studies |
|---|---|---|---|---|---|
| **Price of the book in Rs** | 72 | 80 | 60 | 40 | 50 |
| **Quantity of the book** | 5 | 4 | 6 | 3 | 2 |

Here quantity shows importance. Find weighted Mean price of the book. (Hint: Calculate Weighted Mean).

**Solution:** Weighted Mean = $X_w = (\Sigma x_i\, w_i) / (\Sigma w_i)$

$$= \frac{(72 \times 5) + (80 \times 4) + (60 \times 6) + (40 \times 3) + (50 \times 2)}{5 + 4 + 6 + 3 + 2}$$

$$= \frac{360 + 320 + 360 + 120 + 100}{20} = \frac{1260}{20} = 63 \text{ rupees}$$

Thus, weighted Mean is 63, i.e., Mean price of the books is Rs 63.

### Median:

Median is the middle value of the data. It divides the data into two parts. For finding the median value of a given ungrouped data first we have to arrange the given data in order. If the number of items (n) in a given data is odd, then the middle term will be median, i.e., (n+1)/2 th term.
If the number of items in a given data is even, then mean of the two middle terms will be the median.

**Example 4.** Find the median of the data: 2, 10, 6, 8, 9, 5, 3, 7, 4.
**Solution:**
First of all, given data is arranged in ascending order.
2, 3, 4, 5, 6, 7, 8, 9, 10.

Here the given data consist of 9 terms (9 is an odd number)
Therefore Median = (n+1)/2 th value. Here n = 9.
Hence Median = (9+1)/2 th value = (10/2) th value = 5th value = 6.

**Example 5.** Find the median of the data: 7, 10, 11, 13, 4, 8, 14, 12, 6, 3.
**Solution:** First of all, given data is arranged in ascending order.
3, 4, 6, 7, 8, 10, 11, 12, 13, 14.
In the given data there are 10 terms and it is an even number. So mean of the middle two terms is the median, i.e.,
Median = (10 + 8) / 2 = 18 / 2 = 9.

**Mode:**
Mode is that value which occurs maximum number of times in a group of data. It is that value which has the greatest frequency in a given grouped data.

**Example 6.** The marks obtained by a group of 12 students are given as 70, 50, 40, 65, 45, 65, 70, 65, 50, 45, 70, 65. Find the mode of the data.
**Solution:** Given data is arranged in ascending order.

$$40,\ \underbrace{45, 45}_{2\text{ times}},\ \underbrace{50, 50}_{2\text{ times}},\ \underbrace{65, 65, 65, 65}_{4\text{ times}},\ \underbrace{70, 70, 70}_{3\text{ times}}$$

In the given data 65 comes four times, whereas other numbers occur less number of times.
Hence, 65 marks is the mode of the marks obtained by students.

## EXERCISE 12.2

**Find the mean value of the following data:**
**(i)** 3, 8, 5, 4, 6, 0, 7, 1, 2
**(ii)** 11, 13, 15, 7, 9, 1, 3, 5
**(iii)** 11, 16, 13, 17, 10, 15, 18, 14, 12
**(iv)** 4, 16, 32, 8, 40, 12, 20, 28, 24, 36
**(v)** 1.2, 2.4, 3.6, 4.8, 5.1, 6.3, 7.5, 8.7

**Find the value of required terms in the following:**
**(i)** X = 12, Σx = 132, n = ?
**(ii)** n = 15, Σx = 75, X = ?
**(iii)** Σx = 96, X = 8, n = ?
**(iv)** X = 13, n = 11, Σx = ?

**1. Find weighted mean of the following.**

**(I)**

| Number (x) | 10 | 20 | 30 | 40 | 50 | 60 | 70 |
|---|---|---|---|---|---|---|---|
| Weight (w) | 1 | 2 | 1 | 1 | 2 | 2 | 1 |

(ii)

| Number (x) | 0 | 2 | 4 | 6 | 8 | 10 | 12 |
|---|---|---|---|---|---|---|---|
| Weight (w) | 1 | 2 | 3 | 4 | 5 | 3 | 2 |

**2. Find median of the following:**

**(i)** 17, 19, 11, 5, 3, 7, 9, 13, 15

**(ii)** 45, 5, 15, 20, 35, 40, 10, 25, 30

**(iii)** 22, 2, 20, 4, 18, 6, 16, 8, 14, 10, 12

**(iv)** 35, 21, 49, 77, 45, 27, 33, 55, 63, 81, 99

**(v)** 39, 78, 91, 75, 104, 105, 19, 29, 99, 79, 89

**(vi)** 2.5, 3.5, 1.5, 4.5, 5.5, 5.0, 9.9, 9.5, 10.5, 11.5, 6.5, 7.5, 5.5

**3. Find mode of the following:**

**(i)** 14, 21, 27, 14, 28, 29

**(ii)** 24, 31, 37, 24, 38, 39

**(iii)** 23, 29, 41, 23, 29, 77, 55, 29

**(iv)** 20, 26, 38, 20, 26, 74, 52, 26

**(v)** 71, 51, 81, 71, 51, 57, 71, 81, 93, 51, 71

**Fill in the blanks:**

**(i)** 3, 3, 4, 4, 5, 8, 7, 8, ________________, 9, if mode is 8.

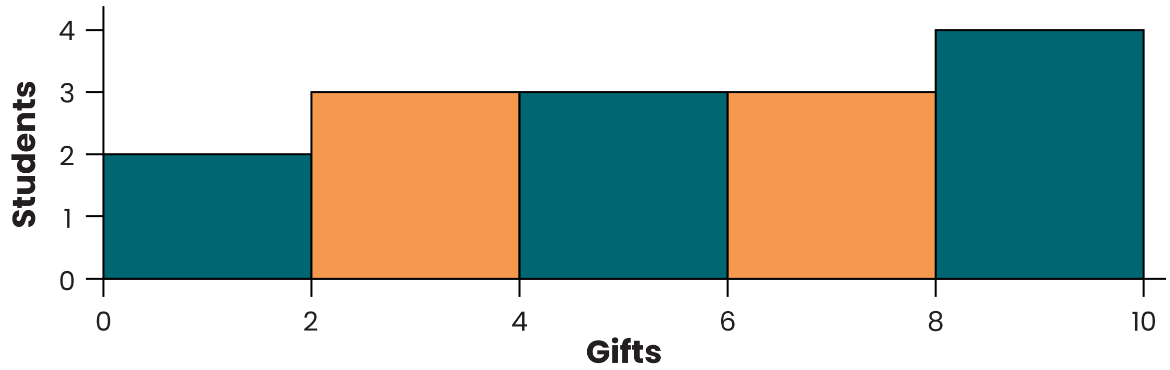
**(ii)** 6, 9, 7, 10, ________________, 4, if mean is 8.

## ACTIVITIES

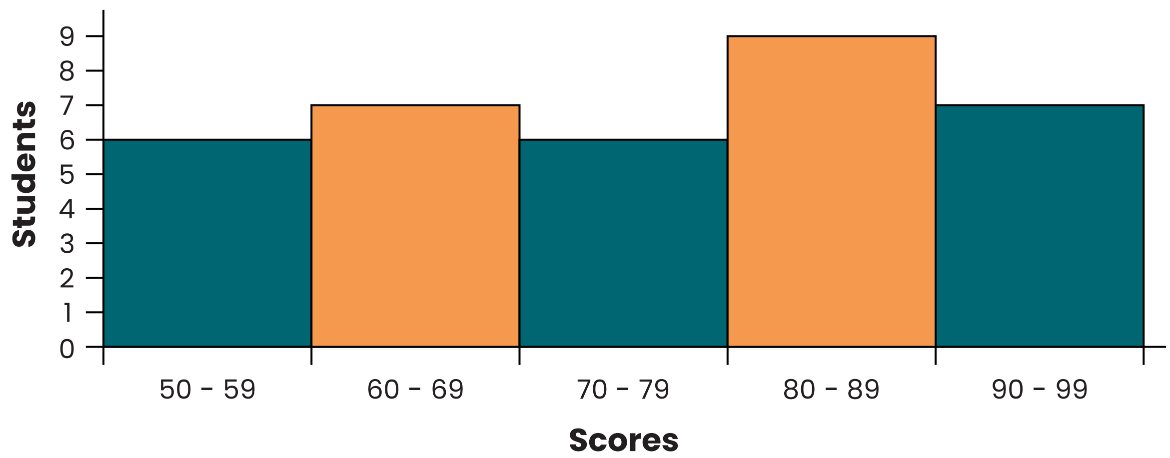
**Q. Find the mean, median and mode for each set of numbers.**

| Given | Mean | Median | Mode |
|---|---|---|---|
| **a.** 4, 4, 3, 9, 5 | | | |
| **b.** 8, 4, 5, 8, 5, 4, 8 | | | |
| **c.** 10, 13, 11, 13, 13 | | | |
| **d.** 5, 12, 13, 16, 12, 14, 12, 5, 10 | | | |
| **e.** 15, 18, 19, 20, 22, 24, 22 | | | |
| **f.** 42, 34, 36, 24, 34 | | | |

**Q. Answer the questions.**




**1)** How many students received between 0 and 2 gifts?
**2)** How many students are represented in this histogram?
**3)** If a student received 6 gifts which bar would they be added to?
**4)** Most students received between _________ and _________ gifts.




**5)** How many students are represented in this histogram?
**6)** How many students scored between a 50 and 60?
**7)** Most students scored between a _________ and _________.

**Q. Solve the following:**

**1)** 82, 23, 59, 94, 70, 26, 32, 83, 87, 94, 32
Mean _________ Median _________ Mode _________

**2)** 83, 93, 77, 33, 62, 28, 23
Mean _________ Median _________ Mode _________

**3)** 31, 92, 25, 69, 80, 31, 29
Mean _________ Median _________ Mode _________

**4)** 67, 70, 49, 95, 40, 97, 62, 64, 42
Mean _________ Median _________ Mode _________

**5)** 86, 13, 60, 85, 61, 97, 30, 98, 79, 52, 18
Mean _________ Median _________ Mode _________

**6)** 24, 22, 32, 59, 99, 59, 76, 83, 21, 95, 57
Mean ________ Median ________ Mode ________
**7)** 21, 62, 66, 66, 79, 28, 63, 48, 59, 94, 19
Mean ________ Median ________ Mode ________
**8)** 83, 23, 24, 71, 52, 62, 63
Mean ________ Median ________ Mode ________
**9)** 63, 25, 43, 28, 72, 61, 45, 46, 13
Mean ________ Median ________ Mode ________
**10)** 53, 44, 10, 45, 59, 97, 77
Mean ________ Median ________ Mode ________

**Mean, weighted mean, median and mode:**

**Example 1.** The weights of 20 students (in kilograms) are given below: 60, 61, 67, 68, 59, 58, 55, 58, 59, 60, 61, 66, 65, 64, 63, 60, 59, 58, 56.
**Find:**
**(I)** The mean weight of students.
**(ii)** The mode or modes of weight of students.
**(iii)** The median weight of students.
**Solution:** First we have to arrange the given data in order, say in descending order:

68, 67, 66, 65, 64, 63, 62, $\underbrace{61, 61}_{\text{2 times}}$, $\underbrace{\boxed{60, 60, 60}}_{\text{3 times}}$, $\underbrace{59, 59, 59}_{\text{3 times}}$, $\underbrace{58, 58, 58}_{\text{3 times}}$, $\underbrace{56, 56}_{\text{2 times}}$

**(i)** Sum = 68 + 67 + 66 + 65 + 64 + 63 + 62 + 2(61) + 3(60) + 3(59) + 3(58) + 2(56) = 1220
∴ X = Σx / n = 1220 / 20 = 61.
Thus, mean weight of students is 61 Kg.

**(ii)** In the given data, there are three modes (60kg, 59kg, and 58kg) because each of the numbers 60, 59, and 58 occur three times.

**(iii)** There are 20 numbers in the data.
Hence, the median is the mean of the 10th and 11th numbers in the data.
i.e., Median = (60 + 60) / 2 = 60.
Thus, median weight of students is 60Kg.

## EXERCISE 12.3

**Q1.** A person purchased the following vegetable items:

| S. No | Vegetable Items | Quantity (in Kg) | Rate/Cost Per Kg (in Rs) |
|---|---|---|---|
| **1.** | Potatoes | 12 Kg | Rs 25 |
| **2.** | Onions | 8 Kg | Rs 30 |
| **3.** | Tomatoes | 5 Kg | Rs 60 |
| **4.** | Carrots | 3 Kg | Rs 40 |
| **5.** | Lady Fingers | 4 Kg | Rs 80 |

Find weighted mean cost of vegetable items per Kg.

**Q2.** The heights of 20 students (in centimetres) of Class VIII are given below:
130, 135, 140, 145, 142, 141, 140, 135, 136, 137, 132, 134, 136, 138, 140, 142, 144, 143, 139, 138.
Compute the mean, mode and median value of height of students.
Is Mean ≥ Median ≥ Mode?

**Q3.** Twenty crates of Sindhri mangoes were sent to Sukkur from Hyderabad. The number of rotten mangoes in each crate, respectively, is given below:
0, 2, 1, 0, 3, 2, 1, 0, 4, 1, 2, 3, 0, 2, 3, 4, 1, 2, 3, 3.
Find the mean, mode and median number of rotten mangoes per crate.
Is Mean ≥ Median ≥ Mode?

**Q4.** Forty-five students appeared in Science and Mathematics test. The number of marks obtained by them is given below.

| No. of Marks Obtained | 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 |
|---|---|---|---|---|---|---|---|---|---|
| No. of Students in Maths | 5 | 4 | 7 | 8 | 6 | 3 | 4 | 5 | 3 |
| No. of Students in Science | 4 | 4 | 8 | 8 | 5 | 3 | 2 | 7 | 4 |

Find the weighted mean of the marks obtained in each subject. In which subject did the performance of students remain better? (Number of students are weights).

## ACTIVITIES

**Find the mean, median and mode of each word problem. Show your work and write your answer in the space provided.**

**1.** The school nurse recorded the height in inches of eight Grade 5 students: 50, 52, 51, 56, 52, 57, 60, 62.

Mean: ______________ Median: ______________ Mode: ______________

**2.** Bryan had his workout routine and tracked the time he had each day for a 1 week-period. The time in minutes were as follows: 45, 60, 50, 55, 40, 75, 60.

Mean: ______________ Median: ______________ Mode: ______________

**3.** Rica and Kathy organized a fund raising event. On the first day, they collected the following amounts: $14, $15, $30, $25, $20, $18, $18, $12, $22.

Mean: ______________ Median: ______________ Mode: ______________